SMART INDIA HACKATHON 2025

- **Problem Statement ID – 25249**
- **Problem Statement Title-** Development of a floating sensor based system for automatic reading and transmission of water levels in the rivers.
- **Theme-** Disaster Management
- **PS Category-** Hardware
- **Team ID- 68904**
- **Team Name:** Team AgriNova

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Proposed Solution

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The proposed solution is an **automated floating sensor system** that **continuously monitors** river water quality and hydrology. It utilizes **solar-powered electronics, wireless data transfer, and multiple probes (for depth, pH, temperature, and GPS)** to ensure real-time depth measurement, reliable tracking and early flood alerts . Deployment is simple, scalable, and enables proactive environmental management through **instant alerts and remote data access**.Manual maintenance is **minimal**, as the sealed enclosure and solar charging provide **long-term**, unattended operation, supporting efficient, **data-driven decision-making** for water resource managers.

How it addresses the problem:

- Multi-Sensor system (ultrasonic, radar, and pressure) for **redundant measurement** ,ensuring accuracy even when individual sensors face **interference**.
- **Consistent data**(5min interval) ensures **detection** of sudden surges or flash floods and **emergency alerts** are raised.
- In high-turbidity regions, a **low drag hydro-turbine** can optionally supplement **solar power**. **Lithium graded batteries** can provide backup for 30days.
- Engineered with **IP68 rated material** for waterproofing and **ABS plastic** for durability(debris impact and extreme weather)
- Robust **ergonomic shape** to tackle resistance to flow in turbulent waters.
- Powered by energy efficient **Cortex-M micro controller**, which handles data logging and filtering.

Innovation & Uniqueness

- **Machine learning** analyzes sensor and rainfall data to predict water levels **24-48 hours ahead**, triggering **early warnings** for timely evacuation.
- **Self-sufficient module** requiring **minimal** human intervention and lasting up to 5 years.
- **GSM/GPRS module** running on **LoRa communication** is used for real-time data transmission, working in even the most **remote areas**
- **High-precision GPS**, enabling the system to map hydrological conditions dynamically and trace the **flow patterns of the river current.**
- Usage of **temperature** and **pH** sensors along with depth sensor to create a comprehensive **multi-parameter river health monitoring platform.**

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TECHNICAL APPROACH

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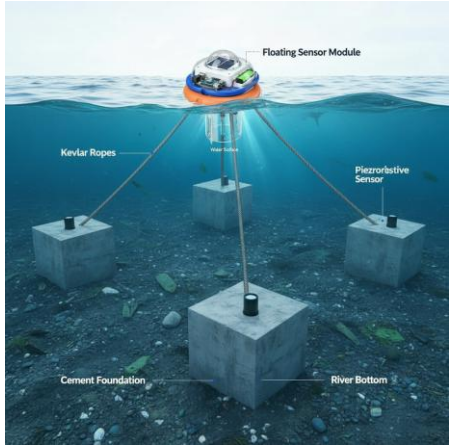
SMART FLOATING WATER ANALYZER

SENSE...DETECT...RESPOND!!

- Ultrasonic Sensors, such as the **Maxbotix MB7040**, measure water level by emitting 40–60 kHz sound waves. This provides **±1 cm accuracy** over **0.3–5 m**, ensuring reliable monitoring in challenging environmental conditions.
- The **Pressure Sensor Array** uses piezoresistive sensing, where diaphragm deformation changes resistance to measure pressure accurately. It's submersible, temperature-compensated, and offers **±0.05% FS accuracy**.
- The **Data Acquisition Unit** (DAU) is driven by a **Cortex-M microcontroller**, efficiently manages **sensor inputs**, **data logging**, and **signal filtering**. DAU performs **accurate data collection** and analysis while supporting scheduled wake-up cycles for optimized power management in continuous monitoring.
- The **ESP32** serves as the control and communication center for the module, linking to **multiple sensors (level, flow, quality) via ADC and GPIO pins**. Its dual-core processor runs **noise filtering and data compression** software at set intervals, integrated Wi-Fi and Bluetooth transmit data over **LoRa WAN** or cellular networks. The **Ultra-Low-Power co-processor** handles basic tasks during deep sleep, greatly increasing battery life in remote locations.



- Proposed solution to prevent the module from drifting along the river current. The structure comes with **Kevlar ropes** which **prevent movement horizontally** and **provides enough stability vertically.**

Step 01 → Step 02 → Step 03 → Step 04 → Step 05

MEASUREMENT

Sensor collect high-resolution data (≥3 parameters: level, flow, quality) at **5-minute intervals**, ensuring **continuous** monitoring of river conditions.

PROCESSING

Microprocessors perform **noise filtration** and **data compression** to optimize data integrity and minimize packet size, conserving power and transmission bandwidth.

TRANSMISSION

Compressed data packets are sent via energy-efficient **LoRaWAN** or **cellular (LTE-M/NB-IoT)** networks to a central gateway, depending on the site's connectivity.

CLOUD INGESTION

Gateways securely push the aggregated data to **robust cloud platforms (Azure/AWS)** for scalable storage and centralized management.

REAL-TIME INSIGHTS

Data is translated into user dashboards, feeds predictive models (e.g., flood forecasting), and powers **automatic alert systems** to enable **rapid, data-driven decision-making**

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FEASIBILITY AND VIABILITY

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Tailored Sensing Solutions for Every Terrain

- ULTRASONIC SENSOR

  Best for: Calm to moderate water conditions

- RADAR SENSOR

  Best for: **Turbulent** water, **extreme** weather zones

- PRESSURE SENSOR

  Best for: **Deep water**, permanent installations

DIRECT COST REDUCTIONS

- **Manual inspection elimination**: Less human intervention, less labour cost.
- **Insurance premium reductions**: Demonstrable risk mitigation lowers premiums 10-15%
- **Infrastructure lifecycle extension**: Early detection of stress prevents catastrophic failures

WHY US??

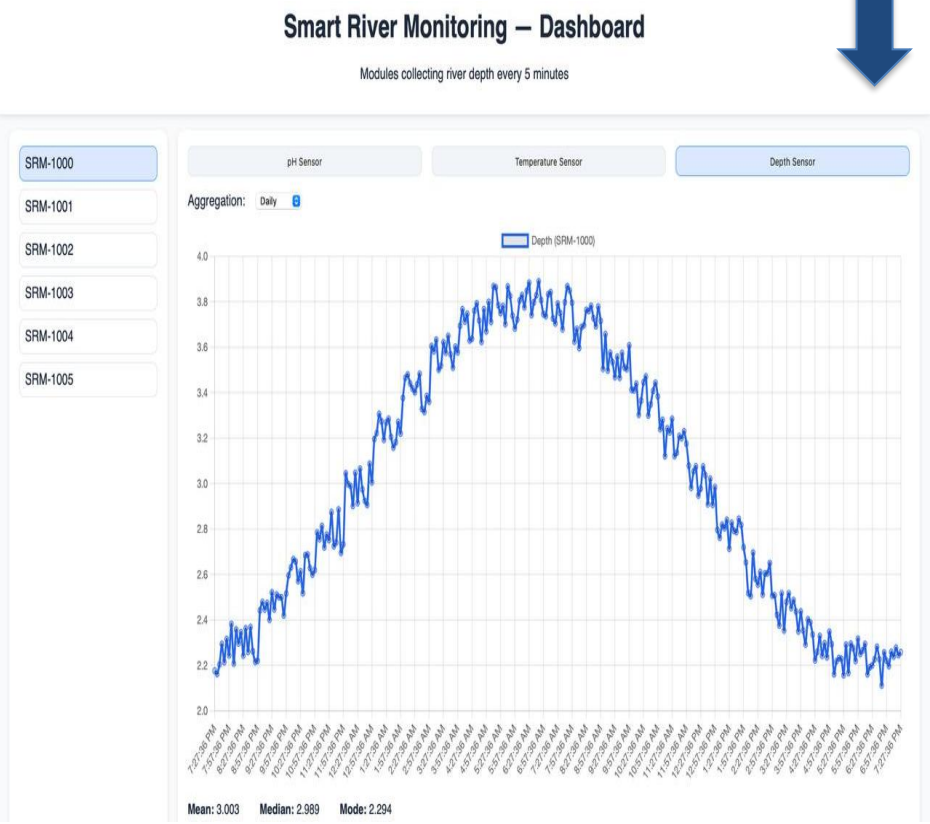
- **The communication aspect-**

  LoRa- For areas with remote access || **LTE-M**- For areas which are well connected|| **Satellite**- For areas which are extremely out of bounds with even natural terrestrial networks unavailable.
- Optimized sleep cycles with **ultra-low power consumption** extends lifespan.
- This is a **self-sustained** module with minimal maintenance .
- **Anti-theft with GPS** providing real time data and security against tampering
- Monitors low water marks in real time, to ensure **water regulations and irrigation scheduling**

Snippets of Our App Working

Copy the link to use for brief working prototype

https://smart-river-monitoring-dashboard.netlify.app/



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IMPACT AND BENEFITS

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Enhanced Flood Prediction

High-frequency data feeds into models, enabling earlier warnings and better evacuation planning to minimize property damage and loss of life

Ecosystem Health Monitoring

Beyond levels, the sensors monitor key water quality indicators (pH, temperature).

LoRa-GSM COMMUNICATION

The combined LoRa–GSM system enables long-range, low-power communication between multiple floating sensors and a central gateway, while GSM/GPRS transmits the collected data to the cloud in real time. This hybrid setup provides wide coverage, low cost, and high scalability, ensuring continuous monitoring of water level

Dynamic Drought And Irrigation Management

Monitor low water marks in real time to enforce water usage regulations and optimize irrigation scheduling for agriculture.

SCALABILITY

The system is scalable, allowing addition of sensors to the LoRa–GSM network for wider coverage. Its low power use, cloud-based data handling, and modular design enable simple, low-cost expansion.

ENERGY EFFICIENCY

The module is low-power, using a DAU sleep cycle that minimizes energy use. The power management allows up to 30 days of battery backup, ensuring reliable operation even in remote locations.

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RESEARCH AND REFERENCES

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- Hydrostatic Pressure Measurement with Piezoresistive Level Sensors https://www.pmc1.com/resources/technical-articles/hydrostatic-pressure-measurement-with-piezoresistive-level-sensors/

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- Early Flood Detection Systems Using IoT https://nevonprojects.com/early-flood-detection/

References